Republic of the Philippines
North Eastern Mindanao State University
Cantilan Campus
Pag-antayan, Cantilan Surigao del Sur 8317
Mobile Computing 1 (CS 325)

A Case Study of

"Enhancing Emergency Communication and Response Coordination through a Smartphone-Based Alert System in Buntalid, Cantilan, Surigao del Sur"

Researcher

Charmie Joy D. Dela Cruz

BSCS-3B

Presented to:

Lloyd Vincent P. Cubero

Instructor 1

Republic of the Philippines
North Eastern Mindanao State University
Cantilan Campus
Pag-antayan, Cantilan Surigao del Sur 8317
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Abstract

This case study examines the design, development, and evaluation of a smartphone-based emergency alert and disaster monitoring system developed to improve disaster communication, evacuation coordination, and community safety in Buntalid, a rural barangay in Cantilan, Surigao del Sur, Philippines. The study addresses persistent challenges in emergency information dissemination, including delayed warnings, limited access to localized hazard updates, weak coordination among residents, responders, and local government units, and the lack of real-time monitoring systems during disasters. These issues are more evident in rural and coastal communities such as Buntalid, where communication infrastructure and immediate access to emergency information remain limited. Despite existing disaster management initiatives, many residents still experience difficulties receiving timely evacuation notices and location-specific alerts during heavy rainfall, flooding, landslides, and other natural hazards. Once hazardous conditions are detected, the system automatically sends emergency alerts and evacuation notifications to all registered residents of Buntalid through their mobile phones.

The application includes a one-tap emergency alert feature, real-time push notifications, GPS-based location tracking, barangay-level hazard mapping, evacuation center navigation, and two-way communication channels between residents and emergency responders. The proposed system is also

designed to identify, monitor, and display disaster-prone areas through an interactive map that shows locations vulnerable to flooding, landslides, storm surges, and other hazards. During emergency situations, the map can display affected locations, evacuation routes, safe zones, evacuation centers, and the real-time locations of residents requiring assistance. This feature enhances situational awareness and allows responders and local government units to coordinate rescue and evacuation operations more effectively and efficiently.

Findings indicate that the proposed system significantly improves the timeliness, accuracy, and accessibility of emergency information dissemination within the community. Participants reported improved disaster awareness, faster incident reporting, and more organized coordination between civilians and emergency responders during simulated disaster scenarios. Residents also expressed that automatic evacuation alerts sent directly to registered mobile phones provided a greater sense of safety and preparedness, especially during periods of continuous heavy rainfall. The one-tap alert feature was found to be highly effective for rapid communication, even among users with limited technical knowledge.

However, the study also identified several challenges, including internet dependency, device compatibility issues, inconsistent network coverage in geographically isolated areas, and the need for continuous maintenance of sensors and monitoring equipment. To improve system reliability and usability

in low-connectivity environments, the study recommends integrating SMS-based fallback alerts, offline GPS functionality, low-bandwidth optimization, renewable backup power for sensors, and cross-platform mobile support. The study further recommends integrating the system with national disaster management frameworks such as the National Disaster Risk Reduction and Management Council (NDRRMC) and hazard-monitoring platforms such as Project NOAH to improve data accuracy, weather forecasting, and interoperability.

In conclusion, the proposed smartphone-based emergency alert and disaster monitoring system demonstrates strong potential to improve emergency communication, evacuation preparedness, and disaster response coordination in Buntalid, Cantilan, Surigao del Sur, particularly in rural communities where access to immediate and reliable emergency information is limited. By ensuring that all residents are registered in the alert system and by integrating real-time environmental sensors, hazard mapping technologies, and automated evacuation notifications, the system offers a practical, efficient, and scalable solution for strengthening public safety and disaster resilience. Ultimately, this study highlights the importance of technology-driven communication and monitoring systems in supporting faster, safer, and more responsive local disaster management practices.

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Chapter 1

Introduction

Effective communication plays a critical role in ensuring public safety, particularly during emergencies such as floods, landslides, typhoons, accidents, According to Ferrer (2026), the Philippines faces frequent natural hazards, including typhoons, floods, and earthquakes, highlighting the need for effective Disaster Risk Reduction and Management (DRRM) strategies. In resent time, smartphones are widely spread and are being used by the majority of Filipino people. It has powerful features such as internet connectivity, GPS, and instant notifications that can be used to improve how emergencies are reported and handled. Instead of relying only on traditional methods, a smartphone-based alert system can provide a faster and more reliable way for people to ask for help.

The timely dissemination of accurate information enables individuals, responders, and local authorities to respond appropriately, thereby minimizing risks, reducing damages, and saving lives. Over the years, communication methods have evolved from traditional systems such as sirens, radio broadcasts, and manual reporting to modern digital platforms. Nowadays, with the innovated technologies showing up in the markets, people are learning to adjust to it. With that

situation happening, the people are also learning how to gather data quickly, where this data holds significant information that can be transmitted on different communication channels like social media platforms (Cruz, Gabatin, et.al. 2024).

Globally, mobile-enabled technologies have significantly improved disaster management by enhancing emergency communication, preparedness, and response coordination. Integrated mobile alert systems, smartphone applications, and SMS-based notification platforms are increasingly being adopted to disseminate real-time emergency information and support efficient management of disaster situations (Repanovici et al., 2022; Aisyah et al., 2026). These technologies allow authorities to quickly disseminate emergency information to affected populations while enabling citizens to report incidents in real time through mobile platforms. Recent studies have demonstrated that integrating Internet of Things (IoT) sensors, water-level monitoring devices, rainfall sensors, and GPS technologies into mobile-based emergency systems significantly improves disaster monitoring, flood prediction, and early warning capabilities in vulnerable communities (K. Praveen and R. Vadivel, 2024; Kaphungkui, 2024).

In the Philippine setting, emergency communication remains a significant challenge, particularly in rural and geographically isolated barangays where access to reliable communication infrastructure is limited. Buntalid,

Cantilan, Surigao del Sur is one of the communities vulnerable to flooding and heavy rainfall during typhoon seasons. Residents often experience delayed emergency warnings, limited access to real-time disaster information, and weak coordination between community members and emergency responders. During continuous rainfall, rising water levels may not be immediately detected, causing delays in evacuation and disaster response. Furthermore, unstable internet connectivity and limited communication facilities in rural areas make emergency coordination more difficult.

Given these challenges, there is a growing need for an innovative and localized emergency communication system that can provide faster, more reliable, and location-based disaster alerts. This case study, entitled “Enhancing Emergency Communication and Response Coordination through a Smartphone-Based Alert System in Buntalid, Cantilan, Surigao del Sur,” aims to develop a smartphone-based one-tap alert system integrated with flood-monitoring sensors, weather-monitoring sensors, and real-time GPS tracking. The proposed system is specifically designed to address the flooding issues commonly experienced in Buntalid during periods of heavy rainfall and severe weather conditions.

The proposed system will require all residents of Buntalid to register their mobile phone numbers and smartphones in the emergency alert database of the barangay. Through this registration process, residents will automatically

receive emergency notifications, rainfall warnings, evacuation announcements, and disaster alerts directly on their mobile devices. When continuous rainfall is detected and flood levels become dangerous, the system will immediately send alerts instructing residents to evacuate to designated evacuation centers or safe areas. This ensures that all registered residents are informed quickly and can respond immediately during emergency situations.

The purpose of this case study is to examine how a smartphone-based alert system integrated with environmental sensors can improve emergency communication and disaster response coordination within the community. The study focuses on the integration of rain sensors and water-level sensors that can automatically detect heavy rainfall, rising floodwaters, and unsafe weather conditions. Once abnormal environmental conditions are detected, the system will automatically trigger emergency notifications to residents, barangay officials, and emergency responders. Through the use of GPS technology and digital mapping, the system can identify and display areas that are highly vulnerable or prone to disasters such as flooding and landslides. These hazard-prone areas can be viewed through real-time maps and location tracking features, enabling authorities to determine which communities require immediate assistance, monitoring, and evacuation.

The proposed system includes features such as one-tap emergency reporting, automatic weather and flood alerts, GPS-based location tracking,

barangay hazard mapping, and two-way communication between users and emergency responders. The system will also provide visual maps showing flood-prone and high-risk areas within the barangay to help residents and authorities monitor dangerous locations during disasters. To ensure functionality even in low-connectivity areas, the study also recommends integrating SMS-based fallback alerts, offline GPS caching, and low-bandwidth optimization. These features allow emergency messages and location data to be transmitted even without stable internet access, making the system more suitable for rural communities like Buntalid.

Despite advancements in emergency communication technologies, recent studies noted that many disaster management systems still face limitations in integrating real-time sensor data, geospatial visualization, and reliable communication services in rural and underserved communities. Researchers emphasized that inadequate infrastructure, unstable internet connectivity, and limited access to advanced monitoring technologies continue to hinder effective disaster response and information dissemination in vulnerable areas (Nkwunonwo et al., 2023; Saputra et al., 2025). Existing systems often rely heavily on internet connectivity and manual reporting, which may delay emergency response during disasters. This study addresses these gaps by proposing a localized smartphone-based alert system capable of automatically detecting unsafe weather conditions, monitoring flood risks,

generating immediate evacuation warnings, and improving coordination among residents, responders, and local government units.

This study aims to develop and evaluate a smartphone-based alert system integrated with flood-monitoring and weather-monitoring sensors to enhance emergency communication and response coordination in Buntalid, Cantilan, Surigao del Sur. Specifically, it seeks to improve the speed and accuracy of emergency alerts, provide real-time flood monitoring and location-based reporting, identify disaster-prone areas through digital mapping, strengthen coordination among stakeholders, and support disaster preparedness within the community.

This study is anchored on the principle that reliable communication systems play a critical role in strengthening disaster risk reduction and emergency management by improving coordination, preparedness, and response effectiveness (Aldawood & Skinner, 2019; Aljohani, et al. 2023). It is also guided by findings that mobile technologies integrated with IoT-based sensor networks, GPS tracking systems, geospatial mapping tools, and real-time data analytics enhance situational awareness, support faster information processing, and improve the accuracy and efficiency of emergency decision-making (Pandey et al., 2025; Bail et al., 2021).

The conceptual framework of the study follows the Input–Process–Output (IPO) model. The input consists of emergency reports, rainfall data, weather sensor readings, water-level sensor readings, registered mobile user information, and user location data. The process involves sensor monitoring, weather detection, GPS processing, hazard mapping, data validation, and automatic alert generation. The output includes real-time emergency notifications, evacuation alerts, mapped flood-affected and disaster-prone areas, and improved communication and coordination among residents and responders. The expected outcome is a more efficient, reliable, and location-aware emergency communication system that enhances public safety, disaster preparedness, and evacuation response in Buntalid, Cantilan, Surigao del Sur.

This study seeks to answer the following research questions:

1. What are the common challenges affecting emergency communication and response coordination in Buntalid?
2. How can a smartphone-based one-tap alert system improve the timeliness and accuracy of emergency notifications?
3. How can rain sensors and water-level sensors improve early detection and warning during heavy rainfall and flooding?
4. How does location-based mapping help identify flood-prone and disaster-prone areas during emergencies?

5. In what ways can the proposed system improve coordination among residents, responders, and local authorities?

6. How can automatic evacuation alerts help residents move safely to evacuation centers or safe areas during disasters?

7. What are the benefits of integrating SMS fallback and offline GPS features in rural emergency communication systems?

The findings of this study are expected to provide valuable contributions to various stakeholders:

Community Members – The system enables residents to receive real-time flood warnings, weather alerts, and evacuation notices directly through their registered mobile phones, improving safety and disaster preparedness.

Emergency Responders – The system improves situational awareness, enhances coordination, identifies high-risk areas through mapping, and reduces response time during emergencies.

Local Government Units (LGUs**)** – The system supports disaster risk reduction planning, flood monitoring, evacuation management, and communication efficiency in rural communities.

Educational Institutions – The study serves as a reference for future research in mobile systems, sensor technologies, disaster management, geographic information systems, and information technology.

Future Researchers – The study provides a foundation for the development of advanced mobile-based disaster communication, automated alert systems, and sensor-integrated emergency management technologies.

Chapter 2

REVIEW OF RELATED LITERATURES AND STUDIES

This chapter reviews related literature and studies that support the development of the proposed system, “Enhancing Emergency Communication and Response Coordination through a Smartphone-Based Alert System in Buntalid, Cantilan, Surigao del Sur.” It aims to establish a solid theoretical and empirical foundation by examining existing research on emergency communication, mobile alert systems, and disaster response coordination. The review covers both foreign and local studies on smartphone-based emergency technologies, real-time communication, and the integration of tools such as GPS, artificial intelligence, and geographic information systems (GIS). These sources help identify existing gaps and inform the design of a system suited to Buntalid, a rural barangay in Cantilan, Surigao del Sur, where distance from the municipal center, limited infrastructure, and a small population create significant challenges in emergency response and communication.

Foreign Related Literature

According to the research by Tong, Zhou, et.al. (2025) entitled “Mobile applications enhance out-of-hospital cardiac arrest outcomes: a systematic review and meta-analysis”, mobile applications are now widely implemented across the globe. However, most existing studies are either observational or based on randomized controlled trials (RCTs) with limited sample sizes, which may constrain the generalizability of their findings. Results highlighted the

significant potential of mobile applications in enhancing bystander interventions and improving patient out-comes. In the research entitled “Improvement of Emergency Situation Management through an Integrated System Using Mobile Alerts” by R. M. Repanovici, S. Nedelcu, et.al. (2022) stated that case studies on the sequence of steps in emergency situations have demonstrated that the speed of reactions and interactions with rescue forces is crucial. Reducing the time from the onset of the intervention is the most important feature. A critical analysis of existing mobile emergency applications on the market revealed that mobile emergency alerts can reduce the overall call duration by reducing the time for making the call, the period of time until the call is answered, and the time required to obtain personal information, address details, and an event description.

In another literature by Wolniak and Grebski (2023) “The Usage of Smartphone Application in Smart City Development – Public Safety and Emergency Services” The integration of smartphone applications into smart city development for public safety and emergency services is a pivotal step toward creating safer, more efficient, and resilient urban environments. These apps empower citizens and authorities to collaborate effectively, enhancing response times and resource allocation during emergencies. According to A. Ahmed, I. I. Azad, and M. M. F. I. Barno (2025) in their research, “Importance of Intelligent Transportation Systems to Analyze Transportation Safety and Crash Risk in Developing Countries” intelligent transportation systems (ITS) integration is not

only a technical advancement within such developing countries through the vehicle-driven approach but also a paradigm shifts towards data-based, proactive and equitable management of the transport system as well as a solution to meet the targets of global road safety under Sustainable Development Goals (SDGs). ITS is not just a futuristic concept for affluent nations, other developing nations now urgently find themselves in need of the road safety crises.

Furthermore, in research entitled “Earthquake Wrangler: Leveraging IOS Technology for Earthquake Detection and Early Warning application to Enhance Public Safety” by Luis F. Salome (2025) Earthquakes are devastating natural phenomena and generate secondary hazards such as tsunamis, landslides and fires. Their catastrophic impacts span both developed nations including the United States, Japan, Turkey, and Italy and other developing countries. While some regions have adopted advanced early warning systems based on seismic hazard models and strong ground motion analysis, others lack the technical and economic capacity to deploy these life saving measures. According to A. Hauri, A. Kohler, and B. Scharte (2022) in their article entitled “A Comparative Assessment of Mobile Device-Based Multi-Hazard Warnings: Saving Lives through Public Alerts in Europe”, while there are many different forms of public warning systems, the ones that use means of electronic communications services offer several advantages, such as increased reachability, as a large part of the population always has a mobile with them.

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Local Related Literature

In the research of Riel M. Rivas (2026) entitled “Barangay Information Management System with Automated Mobile Messaging for Improved Community Engagement” stated that an automated mobile messaging feature was incorporated to facilitate real-time communication. The system enables barangay officials to send announcements, reminders, and urgent notifications directly to residents’ mobile devices. Non-emergency messages are recorded without triggering further action. However, emergency-related messages are analyzed using a rule-based method to classify their nature and determine the appropriate response. According to Aisyah, Rahman, et.al. (2026) in their research “Mobile-Enabled Technologies in Disaster Management: A Systematic Literature Review” Future efforts must focus on enhancing accessibility, addressing user needs, expanding features, and fostering stakeholder collaboration to improve the effectiveness of mobile technologies in disaster preparedness, response, and recovery

Additionally, according to J. Yaneza, J. Bartolome, et.al. (2025) in their study entitled “Online Emergency Response Management System for City Disaster Risk Reduction Management Office of Calamba City, Laguna” One of the critical challenges in disaster-prone areas is the need for effective tools and strategies to mitigate risks and protect vulnerable communities. The system for City Disaster Risk Reduction Management Office of the Calamba City, Laguna

was developed to address these challenges by integrating Geographic Information System (GIS) technology with machine learning. In an article entitled “Mobile Application for Incident Reporting” by M. Ignaco (2021), reporting an incident always depends on self-reporting to the nearest law enforcer's office or calling a channel using a mobile phone. 911 is the National Emergency hotline to get assistance when an emergency occurs. However, the emergency hotline operated by the Emergency Network Philippines (ENP), cannot retrieve the reporter's location details immediately.

According to research by D. N. Aisyah, F. M. Rahman, and H. Diva (2026) entitled “Mobile-Enabled Technologies in Disaster Management: A Systematic Literature Review” the review included 26 studies covering 77 mobile apps across 14 countries. Most apps were privately owned (78.26%), supported multiple disaster phases (41.56%), and favored the Android platform (46.67%), with GPS being the most common technology (15.58%). Apps primarily targeted the general public (63.64%) and focused on earthquakes (32.47%) and hurricanes (31.17%). In the research by J. M. Bustillo, G. A. Patrimonio, and J. I. Mateo (2020) entitled “Automated Incident Reporting Management System Using Mobile Technology” stated that The proliferation of mobile technology is evident in the Philippines. Employing ICT, specifically on the mobile technology spectrum, to maintain peace and order and the assurance for a community’s public safety is plausible. The fact that the Philippine National Police (PNP) is constantly facing controversies and

challenges in carrying its mandate, mobile technology can offer acceptable business solutions.

Foreign Related Study

In the study by F. Mowbray, F. Mills, et.al. (2023) “A systematic review of the use of mobile alerting to inform the public about emergencies and the factors that influence the public response”, Successful emergency communication depends upon how rapidly and reliably a message can be disseminated, but also on how people respond to the message that they receive. According to S. Mohammed, D. Yakubu, and A. Ibrahim (2025) in their study entitled “Mobile-Based Emergency Medical Response Systems for Rural Healthcare Access: A Comprehensive Approach to Sustainable Development” The mobile-based EMS application presents a viable, cost-effective solution for improving emergency healthcare delivery in rural areas. The system's offline capabilities and user-friendly interface make it particularly suitable for resource-constrained environments.

In another study by J. C. Escolano, A. R. Caballero, et.al. (2023) entitled “Acceptance of Mobile Application on Disaster Preparedness: Towards Decision Intelligence in Disaster Management” stated that a larger number of respondents is necessary to gain better understanding of acceptance as well as for generalization. Likewise, this work did not present yet the interoperability of the proposed architecture with information systems to

support disaster management developed by government agencies in the Philippines. According to K K. Stephens, N. H. Carlson, et.al. (2023) in their research entitled “Disaster Rescue Communication Using Mobile Devices, Social Media, and Artificial Intelligence”, as AI-infused information technologies are being developed, often by incorporating human expertise, they are making progress toward being an integral partner in disaster response and recovery. The future of research on using ICTs to bridge some of the existing gaps includes considering more methods, expanding to consider cultural variables, and including disaster practitioners on even broader interdisciplinary research teams.

Additional research by Frank Arena (2023) entitled “Artificial Intelligence in Emergency Response Systems” stated that by leveraging machine learning, computer vision, and natural language processing, AI can enhance situational awareness, optimize resource allocation, and improve the efficiency of search-and-rescue operations. Despite its potential, the integration of AI into emergency systems faces challenges related to data privacy, interoperability, and the ethical implications of automated decision-making.

Local Related Study

According to C. S. Saflor, and K. Kudhal (2026) in their study entitled “Emergency Alert and Warning Systems and Their Impact on Sustainable Disaster Preparedness and Awareness in the Philippines: A SEM–ANN

Analysis" Unlike previous studies that rely solely on SEM or descriptive approaches, the combined SEM–ANN framework enables a more comprehensive understanding of both causal relationships and complex behavioral dynamics influencing disaster preparedness. The findings reveal that behavioral intention, system reliability, message clarity, and trust in EAWS substantially affect individuals' preparedness behavior and risk mitigation actions. In a study by K. L. Benigno, E. Pilar, and Y. Sabado-Lao (2026) entitled "Grassroots Governance and Resilience: The Role of Barangay Disaster Communication Amid Political Uncertainty" stated that traditional communication channels, such as barangay announcements and personal networks, remain central, while political uncertainty can indirectly undermine trust and affect message adherence. Youth participation via SK initiatives contributes to information dissemination but requires enhanced resources and training.

In another study by Y. Han, and G. Liu (2026) entitled "The Role of GIS in Disaster Response: Improving Efficiency and Overcoming Challenges" stated that integration of GIS with advanced technologies such as Building Information Modeling (BIM), remote sensing, artificial intelligence (AI), and machine learning has expanded its potential, enabling more accurate predictions, faster decision-making, and optimized resource allocation. Despite its significant contributions, the use of GIS in disaster management faces

challenges, including issues related to data quality, system interoperability, and user capabilities.

According to X. V. Asuncion, G. M. Suarez, et.al. (2025) in their study "Data Management in Local Disaster Risk Reduction-Climate Change Adaptation in the Philippines: Scenarios and Strategies" the scenarios presented in this research provide a general outlook for the Philippines as a whole rather than focusing on the specific context of individual LGUs. If they wish to better adapt these methods to their particular circumstances, conducting more focused case studies or comparative case studies across different localities could be beneficial.

Synthesis

Based on the reviewed studies from 2022 to 2026, it is clear that smartphone technology plays a very important role in improving emergency communication and response coordination. Many foreign studies show that using smartphones for alert systems makes communication faster, more reliable, and more efficient. For example, several researchers found that mobile alert systems can quickly notify responders and authorities, which helps reduce response time during emergencies. Other studies also highlight that when smartphones are combined with technologies like IoT, GPS, cloud systems, and even artificial intelligence, they become more powerful tools for managing disasters and coordinating responses in real time.

Furthermore, the reviewed literature shows that while current emergency communication systems are helpful, they still have limitations in terms of speed, coordination, and integration. Because of this, there is a need to develop a better system that uses smartphones to improve communication and coordination during emergencies. This supports the goal of the present study, which is to enhance emergency response through a more efficient and reliable smartphone-based alert system.

Chapter 3

Methodology

This study utilized a descriptive-developmental research design to determine the effectiveness of a smartphone-based emergency alert system in improving emergency communication, disaster preparedness, and response coordination in Buntalid, Cantilan, Surigao del Sur. The study aimed to identify the common communication problems experienced during emergencies and to develop a mobile-based system capable of providing faster, more reliable, and real-time communication within the community.

The researchers gathered data through a quantitative survey conducted using Google Forms. The respondents of the study consisted of sixty-one (64) residents from the community, including students, workers, emergency responders, and other community members. The survey questionnaire was designed to collect information regarding the respondents' demographic profile, experiences during emergencies and disasters, challenges encountered in emergency communication, and perceptions regarding the effectiveness, usability, accessibility, and preparedness capabilities of the proposed smartphone-based emergency alert system.

The study used purposive sampling in selecting the respondents. Only individuals who were residents of Buntalid and were familiar with smartphone

usage and emergency situations were included in the study. The respondents were selected to provide relevant information regarding the current emergency communication practices and the possible implementation of the proposed system within the community.

The proposed smartphone-based emergency alert system was designed to provide real-time communication between residents and emergency responders during disasters and emergency situations. All residents of Buntalid were intended to register their mobile phone numbers and devices in the system to ensure that they could receive immediate emergency notifications and evacuation alerts. The system was specifically designed to notify residents whenever continuous heavy rainfall, flooding, landslides, or other dangerous weather conditions occur, informing them if evacuation to designated evacuation centers or safe areas is necessary.

The system included features such as one-tap emergency alerts, push notifications, GPS-based location tracking, hazard mapping, real-time location sharing, and two-way communication capabilities. In addition, the application was integrated with environmental and weather monitoring sensors capable of detecting unsafe weather conditions such as excessive rainfall and possible flood risks. Once abnormal weather conditions were detected, the system automatically generated emergency alerts and notifications to warn residents and local authorities.

Furthermore, the application contained a disaster-prone mapping feature that allowed users and emergency responders to identify areas vulnerable to flooding, landslides, and other hazards. Through GPS and mapping technology, residents could view safe zones, evacuation centers, and high-risk locations directly within the application. This feature was intended to improve disaster awareness, preparedness, and evacuation response within the community.

The application was developed for Android smartphones due to their accessibility and widespread use among the respondents. During the development phase, the researchers followed a systematic process that included planning, system analysis, interface design, coding, testing, and evaluation. User-friendly interfaces and simple navigation features were incorporated into the application to ensure that users with limited technical knowledge could easily operate the system during emergency situations.

After the development of the application, the researchers conducted system testing and evaluation to determine the functionality, usability, reliability, and effectiveness of the proposed system. The respondents evaluated the system using structured questionnaires based on different criteria such as communication effectiveness, GPS and location tracking features, weather and sensor alert accuracy, usability and accessibility, disaster preparedness, and emergency response coordination.

The gathered data were analyzed using descriptive statistical tools such as frequency counts, percentages, and weighted mean. Frequency counts and percentages were used to describe the demographic profile of the respondents, while weighted mean was used to determine the level of agreement of the respondents regarding the effectiveness of the proposed smartphone-based emergency alert system. The results were interpreted using a five-point Likert scale with corresponding verbal interpretations ranging from Strongly Disagree to Strongly Agree.

Through this methodology, the researchers were able to assess the communication challenges experienced during emergencies and evaluate the effectiveness of the proposed smartphone-based emergency alert system in enhancing disaster preparedness, improving public safety, providing early warning notifications, and strengthening emergency response coordination within the community of Buntalid, Cantilan, Surigao del Sur.

Result and Discussion

This chapter presents the analysis, interpretation, and discussion of the data gathered from sixty- four (64) respondents regarding the proposed smartphone-based emergency alert system. The findings were organized according to the objectives of the study and were presented using frequency counts, percentages, weighted means, and verbal interpretations based on the responses collected from the Google Form survey.

The purpose of this chapter is to determine the challenges encountered in emergency communication, evaluate the effectiveness of the proposed smartphone-based emergency alert system, assess the usefulness of GPS and location tracking features, determine the usability and accessibility of the application, and examine its contribution to disaster preparedness and coordination within the community.

4.1 Respondents' Demographic Profile

Table 1. Age Distribution of Respondents

Age Bracket	Frequency	Percentage
Below 18	2	3.1%
18–25	52	81.3%
26–35	4	6.3%
36–45	3	4.7%
46–55	3	4.7%
56 and above	0	0%
Total	**64**	**100%**

Discussion

Table 1 shows that most respondents were aged 18–25, with 52 respondents or 81.3% of the total population. This indicates that young adults

are the primary users of smartphones and mobile applications, making them more likely to adopt a smartphone-based emergency alert system.

Respondents aged 26–35 accounted for 6.3%, while those aged 36–45 and 46–55 each represented 4.7%. Only 3.1% were below 18 years old, and none were aged 56 and above.

The findings suggest that younger individuals are more familiar with digital technology and more open to using mobile-based emergency communication systems. Meanwhile, older respondents provided additional insights regarding disaster awareness and communication concerns within the community.

Table 2. Gender of Respondents

Gender	Frequency	Percentage
Male	28	43.8%
Female	34	53.1%
Others	2	3.1%
Total	**64**	**100%**

Discussion

Table 2 presents the gender distribution of the respondents. Female respondents comprised the majority with 34 respondents or 53.1% of the total

population, while male respondents accounted for 28 respondents or 43.8%. In addition, 2 respondents or 3.1% identified as others.

The results indicate that both male and female respondents actively participated in the study, allowing the researchers to gather diverse opinions and experiences regarding the proposed smartphone-based emergency alert system. The slightly higher number of female respondents suggests strong engagement and interest in disaster preparedness and emergency communication. Furthermore, the inclusion of respondents from different gender groups contributed to a more balanced and inclusive evaluation of the system's accessibility, usability, and effectiveness within the community.

Table 3. Occupation of Respondents

Occupation	Frequency	Percentage
Student	56	87.5%
Teacher	0	0%
Government Employee	0	0%
Private Employee	1	1.6%
Farmer/Fisherman	3	4.7%
Business Owner	0	0%
Emergency Responder	1	1.6%
Unemployed	2	3.1%

Others	1	1.6%
Total	**64**	**100%**

Discussion

Table 3 presents the occupation of the respondents. The majority of the respondents were students, with 56 respondents or 87.5% of the total population. Farmer/fisherman respondents accounted for 3 respondents or 4.7%, while unemployed respondents comprised 2 respondents or 3.1%. Private employees, emergency responders, and others each had 1 respondent or 1.6%. Meanwhile, there were no respondents who identified as teachers, government employees, or business owners.

The findings show that the respondents were primarily students, indicating that the study mainly gathered data from individuals who are actively engaged in academic activities and are more likely to be familiar with smartphones and digital technology. The inclusion of respondents from different occupations such as farmers/fishermen, private employees, and emergency responders also contributed diverse insights regarding the accessibility, usability, and importance of the proposed smartphone-based emergency alert system within the community.

Table 4. Civil Status of Respondents

Civil Status	Frequency	Percentage
Single	59	92.2%
Married	5	7.8%
Widowed	0	0%
Separated	0	0%
Total	**64**	**100%**

Discussion

Table 4 presents the civil status distribution of the respondents. Out of 64 respondents, 59 or 92.2% are single, while 5 or 7.8% are married. No respondents identified as widowed or separated.

The data indicate that the majority of the respondents are single individuals, showing a highly dominant percentage compared to married respondents. This implies that the study participants are primarily composed of unmarried individuals, while only a small portion are married. The absence of widowed and separated respondents also shows that these civil status categories were not represented in the study population.

Table 5. Length of Residency in the Community

Years Living in the Community	Frequency	Percentage

Less than 1 year	8	12.5%
1–5 years	11	17.2 %
6–10 years	4	6.3%
More than 10 years	41	64.1%
Total	**64**	**100%**

Discussion

Table 5 presents the length of residency of the respondents in the community. The majority of the respondents, with 41 individuals or 64.1%, have been living in the community for more than 10 years. This indicates that most respondents are long-term residents who are highly familiar with the community environment and local conditions. Meanwhile, 11 respondents or 17.2% have lived in the community for 1–5 years, while 8 respondents or 12.5% have stayed for less than 1 year. The smallest group consists of 4 respondents or 6.3% who have resided in the community for 6–10 years.

The data implies that most of the respondents have extensive experience and knowledge about the community, which may contribute to a better understanding of local issues, emergency situations, and disaster-related concerns. Their long-term residency may also provide more dependable responses regarding the effectiveness and importance of the proposed emergency alert and disaster monitoring system.

Table 6. Smartphone Ownership

Response	Frequency	Percentage
Yes	61	95.3%
No	3	4.7%
Total	**64**	**100%**

Discussion

Table 6 presents the smartphone ownership of the respondents. Out of 64 respondents, 61 or 95.3% reported that they own a smartphone, while only 3 respondents or 4.7% stated that they do not own one. The data shows that smartphone ownership is very high among the respondents. This implies that most participants have access to mobile technology, which may help them receive notifications, alerts, and important information through smartphone-based systems. The results also suggest that the proposed emergency alert system can be effectively utilized by the majority of the community.

Table 7. Type of Internet Connection Usually Used

Response	Frequency	Percentage
Mobile Data	13	20.3%
Wi-Fi	17	26.6%

Response	Frequency	Percentage
Both	34	53.1%
None	0	0%
Total	**64**	**100%**

Discussion

Table 7 presents the type of internet connection usually used by the respondents. The data shows that the majority of respondents, with 34 or 53.1%, use both mobile data and Wi-Fi for internet access. Meanwhile, 17 respondents or 26.6% rely only on Wi-Fi, while 13 respondents or 20.3% use mobile data only. No respondents reported having no internet connection.

The findings indicate that respondents generally have access to internet connectivity through different means. The large percentage of users utilizing both mobile data and Wi-Fi suggests flexibility and continuous access to online services. This implies that the proposed smartphone-based emergency alert system may effectively deliver notifications and updates to users through internet-based communication channels.

Table 8. Frequency of Mobile Application Usage

Usage Frequency	Frequency	Percentage

Very Often	29	45.3%
Often	21	32.8%
Sometimes	13	20.3%
Rarely	1	1.6 %
Never	0	0%
Total	**64**	**100%**

Discussion

Table 8 presents the frequency of mobile application usage among the respondents. The data reveal that the largest group of respondents, consisting of 29 individuals or 45.3%, reported using mobile applications very often. This is followed by 21 respondents or 32.8% who indicated that they often use mobile applications. Additionally, 13 respondents or 20.3% stated that they sometimes use mobile applications, while only 1 respondent or 1.6% reported rarely using them. No respondents selected the option never.

The results show that mobile application usage is highly common among the respondents, as the majority use mobile applications either very often or often. This suggests that respondents are generally familiar with smartphone technology and mobile-based platforms. The low percentage of respondents who rarely use mobile applications further indicates that mobile technology has become part of their regular daily activities.

Table 9. Experience with Emergencies or Disasters

Response	Frequency	Percentage
Yes	62	96.9%
No	2	3.1%
Total	**64**	**100%**

Discussion

Table 9 presents the respondents' experience with emergencies or disasters. Out of 64 respondents, 62 or 96.9% answered "Yes," while only 2 respondents or 3.1% answered "No." The results show that almost all respondents have already encountered emergency or disaster situations. This suggests that the respondents are highly aware of the challenges and risks brought by disasters, including the need for timely information and immediate response. The high percentage of respondents with disaster experience further emphasizes the importance of developing a smartphone-based emergency alert system that can provide fast communication, real-time updates, and improved disaster preparedness within the community.

Table 10. Types of Emergencies Experienced

Type of Emergency	Frequency	Percentage
Flood	26	40.6%
Typhoon	16	25%

Landslide	2	3.1%
Fire	1	1.6%
Earthquake	16	25%
Medical Emergency	2	3.1%
Others	1	1.6%
Total	**64**	**100%**

Discussion

Table 10 presents the types of emergencies experienced by the respondents. Floods were identified as the most commonly experienced emergency, with 26 respondents or 40.6%. This was followed by typhoons and earthquakes, which were both experienced by 16 respondents or 25% each. Meanwhile, landslides and medical emergencies were each reported by 2 respondents or 3.1%, while fire and other emergencies had the lowest frequency with 1 respondent or 1.6% each.

The results reveal that floods are the primary emergency experienced by the community, while typhoons and earthquakes are also significant concerns among the respondents. In contrast, landslides, medical emergencies, fire, and other emergencies were less commonly experienced. These findings show that the community is more exposed to natural disasters, particularly floods, typhoons, and earthquakes.

PART II – SURVEY PROPER

Likert Scale Interpretation

Scale	Range	Verbal Interpretation
5	4.21 – 5.00	Strongly Agree
4	3.41 – 4.20	Agree
3	2.61 – 3.40	Neutral
2	1.81 – 2.60	Disagree
1	1.00 – 1.80	Strongly Disagree

Formula for Weighted Mean

Where:

- $\bar{x}$ = Weighted Mean
- f = Frequency
- x = Scale Value
- N = Total Number of Respondents

The weighted mean was used to determine the level of agreement of the respondents regarding the proposed smartphone-based emergency alert system.

A. Challenges in Emergency Communication

Statement	**Mean**	**Interpretation**
Emergency warnings in the community are sometimes delayed.	3.62	Agree
It is difficult to receive accurate emergency information during disasters	3.48	Agree
Communication between residents and responders is not always effective	3.28	Agree
Limited internet/network signal affects emergency communication	3.76	Agree
Traditional communication methods are insufficient during emergencies	3.48	Agree
Overall Mean	**3.52**	**Agree**

Discussion

Table A presents the respondents' assessment of the challenges in emergency communication. The results show an overall mean of **3.52**, interpreted as Agree, indicating that respondents generally experience

difficulties in communication during emergencies and disasters. Among the indicators, limited internet/network signal affects emergency communication obtained the highest mean of **3.76**, showing that poor connectivity is the most significant challenge encountered by the respondents. This was followed by emergency warnings in the community are sometimes delayed with a mean of 3.62, suggesting that timely dissemination of warnings remains a concern.

In addition, respondents agreed that it is difficult to receive accurate emergency information during disasters and that traditional communication methods are insufficient during emergencies, both with a mean of 3.48. Meanwhile, communication between residents and responders is not always effective received the lowest mean of 3.28, though still interpreted as Agree, indicating that communication gaps between the community and emergency responders still exist. Overall, the findings emphasize the importance of improving emergency communication systems to ensure faster, more accurate, and more reliable information dissemination during disasters.

B. Effectiveness of the Smartphone-Based Emergency Alert System

Statement	**Mean**	**Interpretation**
Smartphone-based systems can improve communication	3.95	Agree

One-tap emergency alerts can help send distress messages quickly	3.40	Agree
Real-time notifications can improve public safety during disasters	3.90	Agree
The system helps residents respond faster during emergencies	3.79	Agree
Mobile alerts are more efficient than manual reporting	3.84	Agree
Overall Mean	**3.77**	**Agree**

Discussion

Table B presents the respondents' assessment of the effectiveness of the smartphone-based emergency alert system. The overall mean of 3.77, interpreted as "Agree," indicates that the respondents generally perceived the system as effective during emergencies and disasters. Among the indicators, "Smartphone-based systems can improve communication" obtained the highest mean of 3.95, showing that respondents strongly believe that smartphone technology can enhance communication during emergency situations.

The respondents also agreed that "Real-time notifications can improve public safety during disasters" with a mean of 3.90, suggesting that timely updates and alerts are important in helping communities stay informed and

safe. In addition, “Mobile alerts are more efficient than manual reporting” received a mean of 3.84, indicating that respondents view mobile-based reporting as a faster and more convenient method compared to traditional processes.

Furthermore, the statement “The system helps residents respond faster during emergencies” obtained a mean of 3.79, which reflects the respondents’ belief that the system can support quicker actions and responses during critical situations. Lastly, “One-tap emergency alerts can help send distress messages quickly” received the lowest mean of 3.40, though still interpreted as “Agree,” indicating that respondents still recognize its usefulness in sending immediate distress signals. Overall, the findings imply that the smartphone-based emergency alert system is considered an effective tool for improving communication, safety, and emergency response within the community.

C. GPS and Location Tracking Features

Statement	**Mean**	**Interpretation**
GPS tracking helps locate affected individuals quickly	3.93	Agree
Location-based mapping improves coordination during emergencies	3.90	Agree

Real-time location sharing is useful during emergencies	3.92	Agree
Hazard mapping identifies high-risk areas	3.95	Agree
GPS-based systems improve situational awareness	3.75	Agree
Overall Mean	**3.89**	**Agree**

Discussion

Table C shows that respondents generally agree on the effectiveness of GPS and location tracking features in emergency situations, with an overall mean of 3.89 interpreted as “Agree.” The highest-rated statement, “Hazard mapping identifies high-risk areas,” obtained a mean of 3.95, indicating that respondents recognize the importance of hazard mapping in identifying dangerous locations and improving disaster preparedness. This was followed by “GPS tracking helps locate affected individuals quickly” with a mean of 3.93 and “Real-time location sharing is useful during emergencies” with a mean of 3.92, showing that respondents value GPS technology for faster response and better communication during emergencies. Meanwhile, “Location-based mapping improves coordination during emergencies” received a mean of 3.90, suggesting that mapping features help responders and communities coordinate effectively. The statement “GPS-based systems improve situational awareness” obtained the lowest mean of 3.75 but was still interpreted as “Agree,” indicating that respondents still consider GPS systems beneficial for

monitoring and managing emergency situations. Overall, the findings suggest that GPS and location tracking features are viewed as useful tools for enhancing emergency preparedness, coordination, and response.

D. System Usability and Accessibility

Statement	**Mean**	**Interpretation**
The application is easy to understand and use	3.68	Agree
The interface should be simple and user-friendly	3.81	Agree
People with limited technical knowledge can use the system easily	3.68	Agree
The system should work even with weak internet connectivity	3.75	Agree
SMS fallback alerts are important in rural communities	3.90	Agree
Overall Mean	**3.76**	**Agree**

Discussion

Table D presents the respondents' assessment of the system's usability and accessibility. The overall mean of 3.76, interpreted as "Agree," indicates that respondents generally perceive the proposed emergency communication system as usable, accessible, and suitable for community use.

Among the statements, “SMS fallback alerts are important in rural communities” obtained the highest mean of 3.90, showing that respondents strongly recognize the importance of having alternative communication methods during emergencies, especially in areas with unstable internet connectivity. The statement “The interface should be simple and user-friendly” followed with a mean of 3.81, indicating that respondents value a clear and easy-to-use interface.

Additionally, the statements “The application is easy to understand and use” and “People with limited technical knowledge can use the system easily” both received a mean of 3.68, while “The system should work even with weak internet connectivity” obtained a mean of 3.75. All statements were interpreted as “Agree,” suggesting that respondents believe the system should be accessible, reliable, and easy to operate for different types of users, particularly during emergency situations.

E. Coordination and Disaster Preparedness

Statement	**Mean**	**Interpretation**
The system can improve coordination between	3.85	Agree

residents and responders		
Faster communication can reduce disaster risks and damages	3.85	Agree
The application can improve community disaster preparedness	3.82	Agree
Real-time communication can save lives during emergencies	3.93	Agree
The proposed system can strengthen	3.93	Agree

public safety in the community		
Overall Mean	**3.87**	**Agree**

Discussion

Table D presents the respondents' evaluation of the system's usability and accessibility. The overall mean of 3.76, interpreted as "Agree," indicates that respondents generally perceive the proposed emergency communication system as user-friendly and accessible.

Among the statements, "SMS fallback alerts are important in rural communities" obtained the highest mean of 3.90, showing that respondents strongly recognize the importance of alternative communication methods, especially in areas with unstable internet connectivity. The statement "The interface should be simple and user-friendly" followed with a mean of 3.81, indicating that respondents value a system that is easy to navigate and understand.

Meanwhile, the statements "The system should work even with weak internet connectivity" received a mean of 3.75, while both "The application is easy to understand and use" and "People with limited technical knowledge can use the system easily" obtained a mean of **3.68**. These results suggest that respondents believe the proposed system should remain accessible, reliable,

and easy to use for all individuals, including those with limited technical skills, particularly during emergency situations.

Recommendations

Based on the findings and conclusions of the study, the following recommendations are proposed:

1. The local government unit and disaster management offices may consider implementing a smartphone-based emergency alert system to improve disaster communication and emergency response.
2. Developers of the proposed application should prioritize user-friendly interfaces and simple navigation to ensure accessibility for all users.
3. SMS fallback alerts and offline notification features should be integrated into the system to maintain communication during weak internet connectivity.
4. GPS tracking and real-time location sharing should be enhanced to improve emergency coordination and rescue operations.
5. Regular disaster preparedness seminars and orientation programs may be conducted to educate residents on the proper use of the emergency alert application.

6. Future researchers may conduct further studies involving a larger number of respondents and additional communities to improve the reliability and effectiveness of the proposed system.

Optional Suggestions from Respondents

The respondents also provided several recommendations for improving the proposed emergency alert system. The most common suggestions include:

1. Improve the speed of emergency notifications and warnings.
2. Ensure that alerts can still function in areas with weak signal or internet connection.
3. Add multilingual support for wider accessibility.
4. Include automatic real-time updates during disasters.
5. Improve the reliability and accuracy of emergency alerts.
6. Develop a simple and easy-to-use interface for all users.

These suggestions indicate that respondents are highly interested in having a reliable, accessible, and efficient emergency communication system that can provide immediate assistance and accurate information during emergencies and disasters.

Conclusion

This study concluded that the proposed smartphone-based emergency alert system has strong potential to improve emergency communication, disaster preparedness, and response coordination in Buntalid, Cantilan, Surigao del Sur. The findings revealed that the system is highly acceptable and beneficial for enhancing public safety, particularly in rural areas that are vulnerable to different natural disasters such as typhoons, floods, landslides, earthquakes, fires, and medical emergencies.

The study further concluded that all residents in Buntalid can be registered in the proposed system using their mobile phone numbers to ensure that they receive emergency alerts and evacuation notices during disaster situations. Through this registration process, residents can automatically receive warnings through notifications, SMS alerts, and real-time announcements whenever continuous heavy rainfall, flooding, or other dangerous conditions occur. This feature will help inform residents when it is already necessary to evacuate to designated evacuation centers or safe areas within the community.

Moreover, the proposed system will integrate weather and environmental sensors capable of monitoring rainfall levels, weather conditions, and possible disaster risks within the area. Once the sensors detect unsafe or critical weather conditions, the system can automatically send alerts to residents, barangay officials, and emergency responders. The system will also

include GPS tracking, hazard mapping, and location-based monitoring to identify and display disaster-prone areas on digital maps. This allows authorities and residents to easily determine which locations are at high risk of flooding, landslides, or other hazards, helping improve evacuation planning and emergency response coordination.

The findings also showed that existing emergency communication methods still face several challenges, including delayed warnings, inaccurate dissemination of information, poor coordination, and unstable internet or network connectivity during disasters. Because of these limitations, respondents strongly agreed that traditional communication methods alone are insufficient during emergencies. The proposed smartphone-based alert system addresses these issues through features such as one-tap emergency alerts, real-time notifications, SMS fallback alerts, GPS location sharing, and hazard monitoring systems that remain functional even in areas with weak internet connectivity.

Furthermore, the results support the idea that mobile technology and sensor-based monitoring systems can significantly enhance disaster risk reduction and community preparedness. The findings are consistent with previous studies emphasizing the importance of real-time communication, GPS technology, automated warning systems, and mobile-based emergency

applications in improving coordination between residents and emergency responders while reducing disaster risks and casualties.

Overall, the study concluded that the proposed smartphone-based emergency alert system can strengthen disaster preparedness, improve emergency communication, provide faster evacuation warnings, and support efficient emergency response coordination in Buntalid. The integration of resident registration, automated sensor detection, real-time alerts, and hazard mapping can greatly contribute to protecting lives, improving public safety, and enhancing community resilience during disasters.

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